Machine Learning
Assignment 6

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I. Written Assignment

1. Is there any classification models solve the parameters and predict decision boundary well without using DNN methods (the distribution of valid vs. invalid data is unknown)?
2. Since the regression model predicted the desired temperature poor with small dataset in Assignment 6. I wondered is there any method to define the loss function such that the error between true value and predicted value decrease with the same scale of dataset?

Ans.

For the first question, I cannot find a good method to decide the decision boundary with parametric model. I think the reason is that the number of parameters in the model is fixed. If the shape of the boundary of valid data is complicated, this method might not well-represent it.

But by [1], SVM is a classifier based on convex optimization which determine the decision boundary in according to margin theory, which could predict the decision boundary well without DNN methods.

In [2], the abstract says, our model is trained using spatial datasets on various attributes in various regions, and predicts values on unseen attributes in unseen regions given a few observed

data. Hence, I think this method might decrease the error with the same scale of dataset in comparison to the regression model we did in Week-6.

II. Program Assignment

II. A. Classification Model Using GDA

- **Data Processing**

  I use the same methods as Assignment to transfer the `xml` file into CSV file. And there are three column in this CSV file, longitude, latitude and feature, respectively.

  If temperature $! = -999$, then feature $= 1$. Otherwise, feature $= 0$.

- **Data Partition**

  Training : 70%

  Validation : 15%

  Test : 15%

- **Notation of GDA**

  GDA model assumes that the conditional distribution of the features given the class label is multivariate normal :

$$p(x \mid y = 0, 1) \sim \mathcal{N}(\mu_k, \Sigma),$$

  where μ_k and Σ are the class-specific mean vector and covariance matrix.

  Using the Bayes' Rule, we can derive the posterior distribution on y given x :

$$p(y \mid x) = \frac{p(x \mid y)p(y)}{p(x)}$$

  and the denominator is given by :

$$p(x) = p(x \mid y = 1)p(y = 1) + p(x \mid y = 0)p(y = 0)$$

Then, we use the training data to solve the parameters in PDF of Gaussian distribution. And use these parameters to predict the test data.

- **Accuracy** I use the method AUC(area under ROC curve) to evaluate the accuracy of the model.
  - AUC $= 1.0 \rightarrow$ Perfect Discriminator
  - AUC $= 0.5 \rightarrow$ Randomly Prediction
  - AUC $< 0.5 \rightarrow$ Model prediction in a reverse way

```
==================================
        Comparison Summary
==================================
QDA Test AUC: 0.9326
==================================
```

Figure 1. AUC of GDA discrimination model.

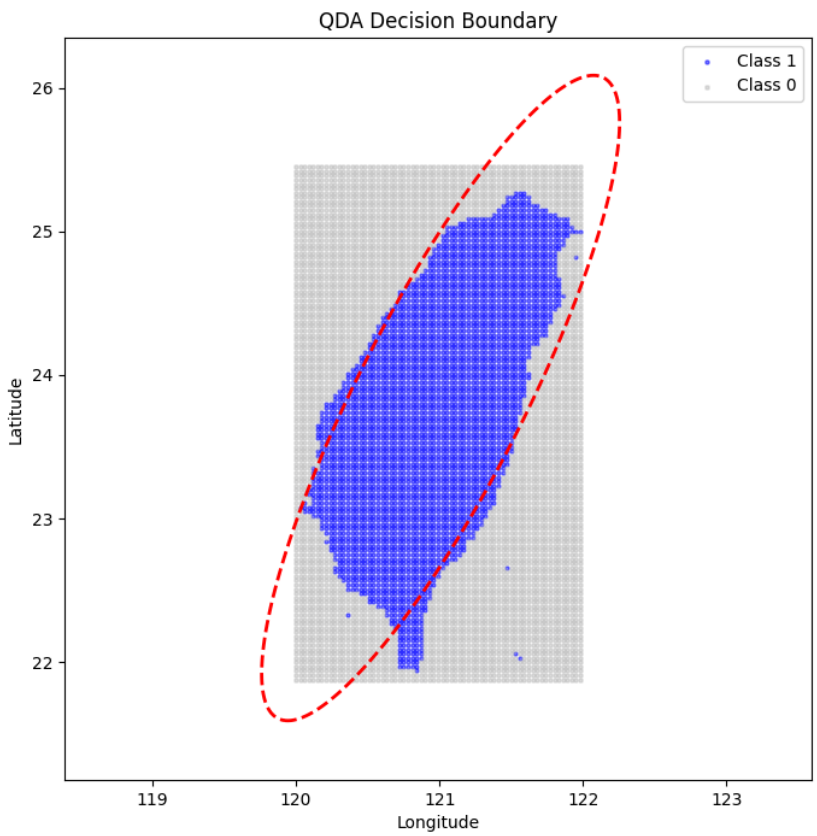

Figure 2. The valid/invalid data points and the decision boundary.

II. B. Regression Model

- **Implementation**

  In `combine_model.py`

  - $C(\vec{x})$: A classification tree model like what I've done in Assignment, but I make it deeper and wider to obtain a better decision boundary then what I've done in Assignment 4. And 70% training, 15% validation and 15% test. And use threshold = 0.5 to improve the accuracy.

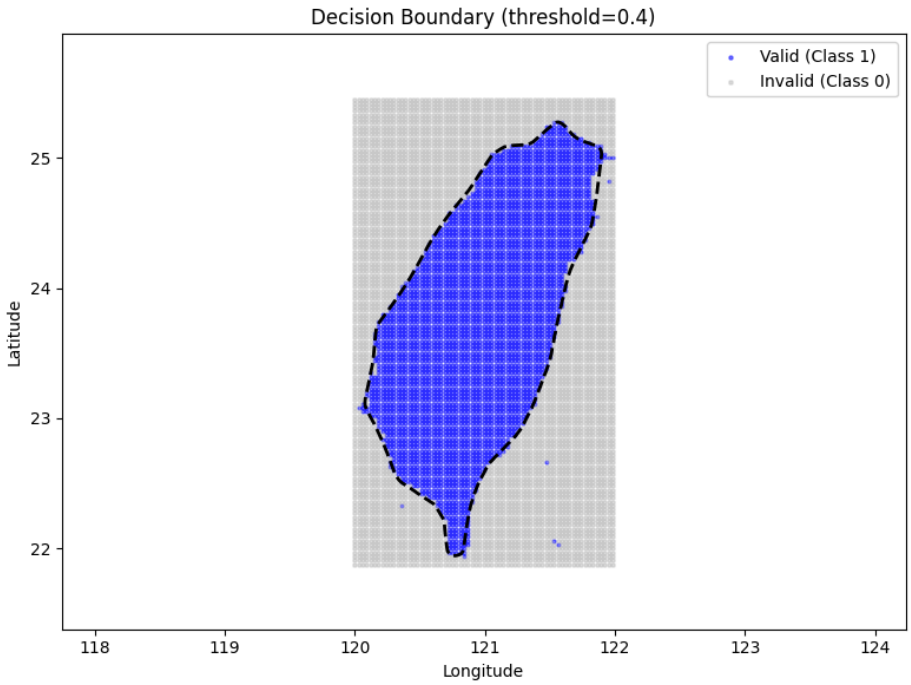

  Figure 3. The decision boundary we obtained by piecewise smooth function model.

  - $R(\vec{x})$: 5 hidden layers, 64 neurons in each layer, and us ReLU activation function, Adam optimizer, MSE loss function. This model only trained on points (temperature $!= -999$).

```
with torch.no_grad():

c_probs = torch.sigmoid(c_model(x).squeeze())

c_pred = (c_probs >= threshold).float()

r_pred = r_model(x).squeeze()

h_pred = c_pred * r_pred + (1 - c_pred) * invalid_value
```

```
    return h_pred
```

The above code is the way I build the combined function, where c_probs, c_pred, r_pred and h_pred are the probability, and prediction obtained from classification model, predicted temperature obtained from regression model and the last is the predicted temperature in this function.

- **Model Evaluation**

This hybrid model did a good job on classification, with AUC = 0.9989, but it did terrible on regression one, I'm thinking if there exist any other way can learn the data better.

```
[Classification] Epoch 1/50 - Train Loss: 0.2942 | Val Loss: 0.1533 | Val AUC: 0.9920
[Classification] Epoch 5/50 - Train Loss: 0.0842 | Val Loss: 0.0933 | Val AUC: 0.9952
[Classification] Epoch 10/50 - Train Loss: 0.0681 | Val Loss: 0.1028 | Val AUC: 0.9948
[Classification] Epoch 15/50 - Train Loss: 0.0668 | Val Loss: 0.0625 | Val AUC: 0.9980
[Classification] Epoch 20/50 - Train Loss: 0.0648 | Val Loss: 0.0707 | Val AUC: 0.9975
[Classification] Epoch 25/50 - Train Loss: 0.0549 | Val Loss: 0.0570 | Val AUC: 0.9982
[Classification] Epoch 30/50 - Train Loss: 0.0551 | Val Loss: 0.0627 | Val AUC: 0.9980
[Classification] Epoch 35/50 - Train Loss: 0.0590 | Val Loss: 0.0659 | Val AUC: 0.9984
[Classification] Epoch 40/50 - Train Loss: 0.0527 | Val Loss: 0.0716 | Val AUC: 0.9974
[Classification] Epoch 45/50 - Train Loss: 0.0489 | Val Loss: 0.0666 | Val AUC: 0.9978
[Classification] Epoch 50/50 - Train Loss: 0.0459 | Val Loss: 0.0435 | Val AUC: 0.9988
[Classification] Test AUC: 0.9989
[Regression] Epoch 1/500 - Train Loss: 492.3066 | Val Loss: 480.6255 | Val MAE: 20.9236
[Regression] Epoch 50/500 - Train Loss: 13.2567 | Val Loss: 14.0261 | Val MAE: 2.8684
[Regression] Epoch 100/500 - Train Loss: 11.4251 | Val Loss: 12.2733 | Val MAE: 2.6437
[Regression] Epoch 150/500 - Train Loss: 9.2846 | Val Loss: 10.8563 | Val MAE: 2.3734
[Regression] Epoch 200/500 - Train Loss: 8.6209 | Val Loss: 10.3193 | Val MAE: 2.2870
[Regression] Epoch 250/500 - Train Loss: 8.2623 | Val Loss: 10.2542 | Val MAE: 2.3893
[Regression] Epoch 300/500 - Train Loss: 8.0199 | Val Loss: 9.7078 | Val MAE: 2.2092
[Regression] Epoch 350/500 - Train Loss: 7.8762 | Val Loss: 9.3977 | Val MAE: 2.1996
[Regression] Epoch 400/500 - Train Loss: 7.8622 | Val Loss: 9.2841 | Val MAE: 2.2322
[Regression] Epoch 450/500 - Train Loss: 7.6491 | Val Loss: 9.0230 | Val MAE: 2.1149
[Regression] Epoch 500/500 - Train Loss: 7.5601 | Val Loss: 9.0407 | Val MAE: 2.1468
[Regression] Test MAE: 10.883°C, RMSE: 15.912°C
```

Figure 4. The train loss, val loss and val AUC during training(classification). The train loss, val loss and val MAE during training(regression).

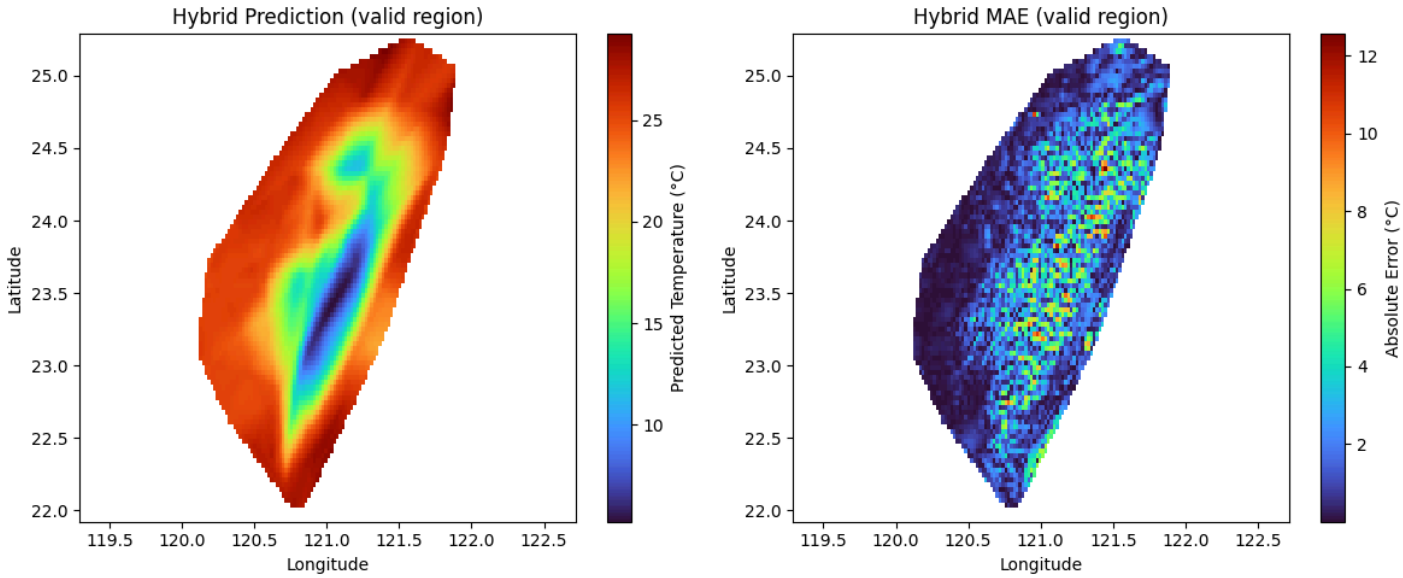

Figure 5. The prediction and the MAE between true and predicted temperature of the piecewise smooth function.

III. Reference

[1] Wikipedia: Support vector machine.

[2] Tomoharu Iwata, Yusuke Tanaka. Few-shot Learning for Spatial Regression.